Ionic Demo

INF 133 Discussion

Concepts Recap

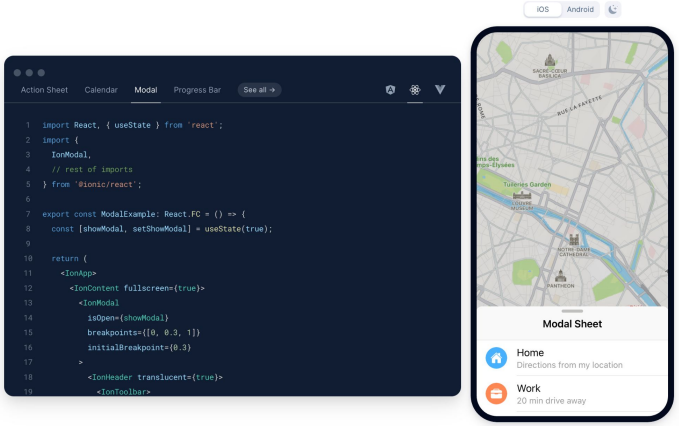
Ionic: “The mobile SDK for the Web.”

- “An open source mobile UI toolkit for building modern, high quality cross-platform mobile apps from a single code base in Angular/Vue/React/Javascript/…”

Benefits

- Cross-platform
- ready-made components for mobile design
- access device APIs through Capacitor.
- deploy to platforms with minimal changes.

(check the website for cross-platform examples!)



Capacitors

- a bridge between **JavaScript code** and **native device** features.
- official plugins (Camera, Local Notifications, Share) + custom native plugins.
- If native APIs aren't available -> still run in browsers using web-based fallbacks
- Native iOS/Android projects remain accessible -> can tweak platform-specific behavior

Capacitor Workflow

1. Write your app as a standard web project inside Ionic (components, routing, logic).
2. Build the web output, which Capacitor syncs into native Android/iOS wrapper projects.
3. Use native IDEs (Xcode, Android Studio) to run on real devices, configure permissions, and publish apps.

This workflow keeps web code separate from native wrappers, making updates easy and predictable.

Getting Started

Installing through npm:

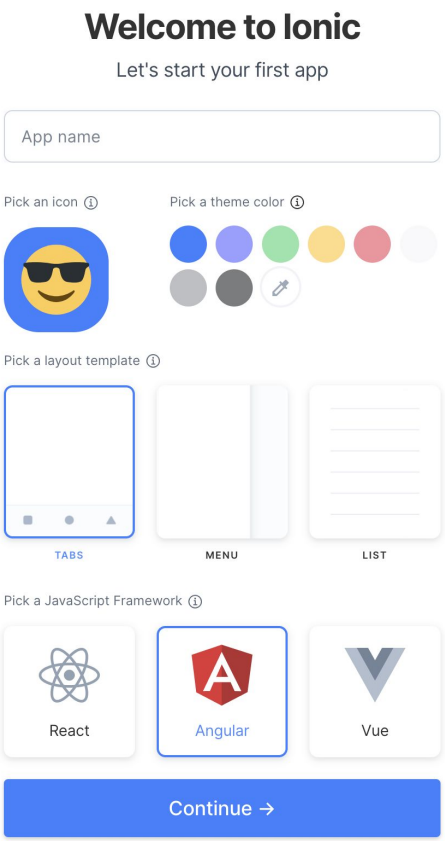
```
$ npm install -g @ionic/cli
```

Starting a new app

```
$ ionic start
```

Running the app

```
$ cd myApp
$ ionic serve
```



The App Wizard

Adding Capacitor, Running on Devices

- For **new** Ionic projects, Capacitor is **already installed** in new Ionic apps by default!
- **Adding capacitors** to existing project:
  - E.g. adding clipboard capacitor
  - # Install the Capacitor Plugins
  - $ npm install @capacitor/camera
  - npx cap sync

```
import { Component } from '@angular/core';
import { Camera, CameraResultType } from
'@capacitor/camera';

@Component({...})
export class CameraComponent{
  public imageSrc: string | undefined = '';

  async takePicture() {
    const image = await Camera.getPhoto({
      quality: 90,
      allowEditing: true,
      resultType: CameraResultType.Uri
    });
    const imageUrl = image.webPath;
    this.imageSrc = imageUrl;
  };
}
```

Using a Native Platform (e.g. iOS)

- Adding the platform
  - npm install @capacitor/ios # you should be able to see a new folder!
  - npx cap add ios
- Opening the iOS Project
  - npx cap open ios # open project in Xcode
  - Or: open ios/App/App.xcworkspace
- Running the app
  - npx cap run ios
- A tip:
  - Check your packages.json and make sure the versions are compatible with each other!
  - If not… you can install the latest version, etc.

Demo Project

Creating a new ionic project and adding UI elements and capacitors

1. Access device information
2. Access and test clipboard
3. Haptics feedback, and how it does not work on macos / web

Try it out on your computer vs phone